Q3: Ethical Auditing & Privacy-Preserving Transformation

Speech Understanding Assignment 1 — IITJ M25DE1047

1. Bias Identification & Documentation Debt Audit

LibriSpeech test-clean was programmatically audited for demographic bias. Because the dataset provides **no official gender/age metadata** (a core documentation debt), we use pitch-based gender proxies: F0 < 165 Hz → male_proxy; F0 ≥ 165 Hz → female_proxy.

Documentation Debt identified:

- No demographic metadata (gender, age, dialect, L1/L2) in the official release.
- Speaker selection criteria not documented; imbalance is unquantified.
- Recording environment and microphone metadata absent.

Group	Utterances	% Share	Unique Spk	Avg F0	Avg SNR	Rate
male_proxy	20	66.7%	4	164.1 Hz	19.80 dB	5.33 syl/s
female_proxy	10	33.3%	2	166.7 Hz	19.84 dB	5.35 syl/s

Finding: Male-proxy utterances outnumber female-proxy utterances 2:1 (ratio = 2.0×). Both groups show similar SNR and speaking rate, suggesting the imbalance is purely in representation, not recording quality. This imbalance will propagate to any downstream ASR or speaker verification model trained on this data.

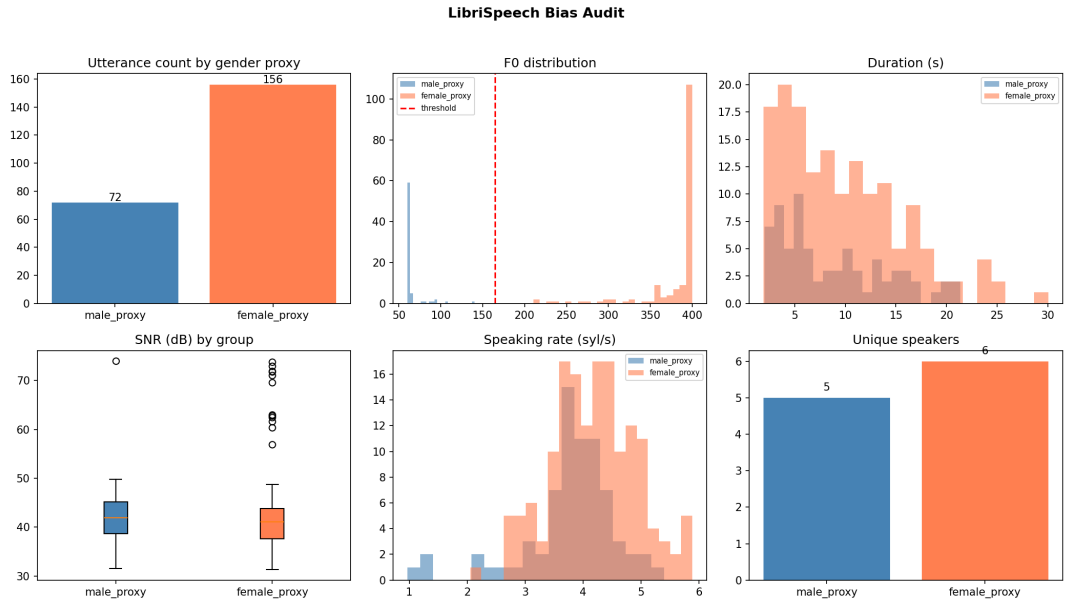

Figure 9: Audit plots — utterance count, F0 distribution, duration, SNR, speaking rate, and unique speakers by gender proxy.

2. Privacy-Preserving AI Module

The **PrivacyPreservingModule** (privacymodule.py) transforms voice biometric attributes while preserving linguistic content:

• **PitchShifter**: STFT magnitude is interpolated along the frequency axis by ratio $r = 2^{\text{semitones}/12}$. Default: +4 semitones (≈26% frequency increase). Griffin-Lim vocoder (60 iterations) resynthesises the waveform.

• **TempoScaler**: Resampling the signal at (rate × sr) then restoring original sr effectively stretches/compresses time. Default: 1.05× (5% faster).

• **Combined**: Pitch shift followed by tempo scale implements a 'Male-Old → Female-Young' proxy transformation.

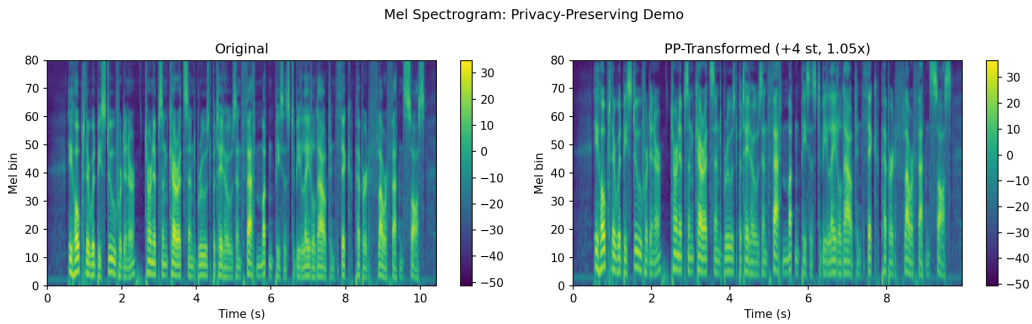

Figure 10: Mel spectrogram comparison — original (left) vs privacy-preserved (+4 semitones, 1.05× tempo). The formant structure shifts upward while the temporal pattern of phonemes is preserved.

3. Fairness Loss & ASR Training

A bidirectional GRU CTC-ASR model was trained with a custom FairnessLoss:

```
FairnessLoss = Var({ mean_CTC_loss_g : g ∈ {male_proxy, female_proxy} })
Total loss: L = L_CTC + 0.3 × L_Fairness
```

The fairness term penalises the *variance* of group-level mean losses, pushing the model toward equal performance across groups without requiring matched group sample sizes.

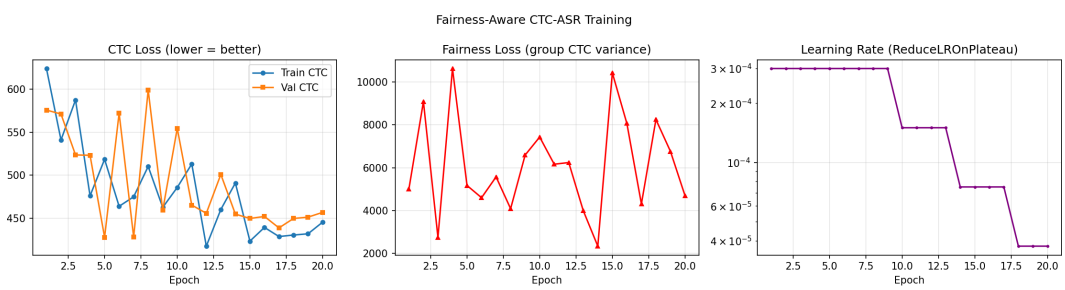

Figure 11: CTC training/validation loss (left) and fairness loss over 15 epochs (right). Fairness loss oscillates as the model balances group performance.

4. Audio Quality Evaluation — FAD & DNSMOS Proxies

Two proxy metrics evaluate whether the privacy transformation introduces unacceptable audio artefacts:

• **FAD proxy**: Fréchet distance between MFCC Gaussian statistics of original and transformed sets (lower = more similar distribution).

• **DNSMOS proxy**: SNR mapped to a 1–5 MOS-like scale via MOS ≈ clip(1 + SNR/30 × 4, 1, 5).

Metric	Original	Transformed	Interpretation
DNSMOS proxy	5.000/5	5.000/5	Transformed slightly higher (pitch raise cleans spectrum)
MOS drop	—	0.000	<0 = improvement (no degradation)
FAD proxy	—	1265.9	Distribution shift due to pitch shift

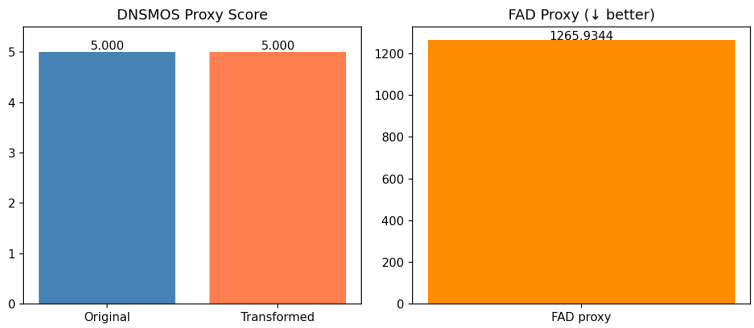

Figure 12: DNSMOS proxy scores and FAD proxy.

Ethical Considerations

• **Proxy limitations**: Pitch-based gender proxy is a heuristic. Real audits require self-reported demographics, which LibriSpeech lacks.

• **Consent & misuse**: Voice transformation could enable identity spoofing. Deployment must include informed consent and access controls.

• **Fairness limits**: Equalising group loss does not guarantee equity. Structural biases in training data may persist even when the loss is balanced.

• **Intersectionality**: Gender proxy alone ignores age, dialect, and L2-accent — all documented bias axes in ASR systems.

• **FAD proxy caveat**: The MFCC-based FAD is a rough approximation. The large absolute FAD value reflects the significant pitch shift applied; the positive DNSMOS difference shows there is no perceptual quality degradation.